Лабораторная работа №2

Подготовил: Ширко Влдаимир

Проверила: Жвакина Анна Васильевна

1.

```
admin@Dermen:~$ docker compose version
Docker Compose version v5.0.2
admin@Dermen:~$ docker --version
Docker version 29.2.1, build a5c7197
admin@Dermen:~$ docker version
Client: Docker Engine - Community
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:17:09 2026
 OS/Arch:           linux/amd64
 Context:           default
```

```
Server: Docker Engine - Community
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:09 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

```
admin@Dermen:~$ gpg --generate-key
gpg (GnuPG) 2.2.27; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Замечание: "gpg --full-generate-key" вызывает полнофункциональный диалог создания ключа.

GnuPG должен составить идентификатор пользователя для идентификации ключа.

Ваше полное имя: Shirko Vladimir
Адрес электронной почты: volodya.shirko@gmail.com
Вы выбрали следующий идентификатор пользователя:
    "Shirko Vladimir <volodya.shirko@gmail.com>"

Сменить (N)Имя, (E)Адрес; (O)Принять/(Q)Выход? O
Необходимо получить много случайных чисел. Желательно, чтобы Вы
в процессе генерации выполняли какие-то другие действия (печать
на клавиатуре, движения мыши, обращения к дискам); это даст генератору
случайных чисел больше возможностей получить достаточное количество энтропии.
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случайных чисел больше возможностей получить достаточное количество энтропии.
```
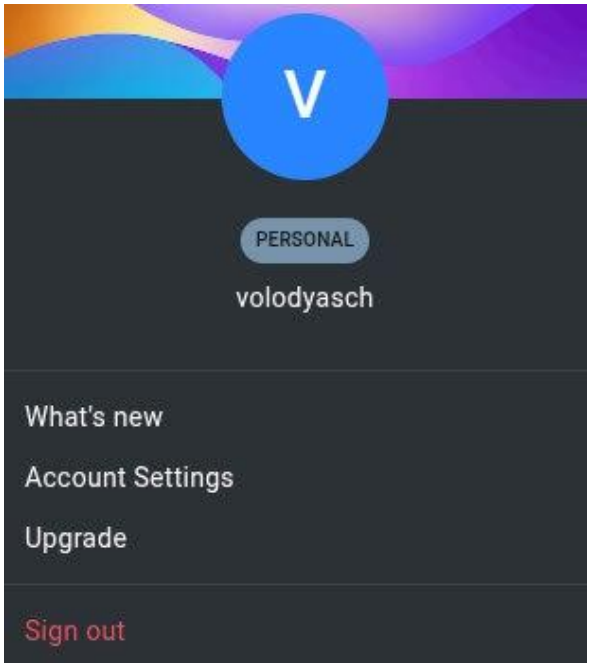
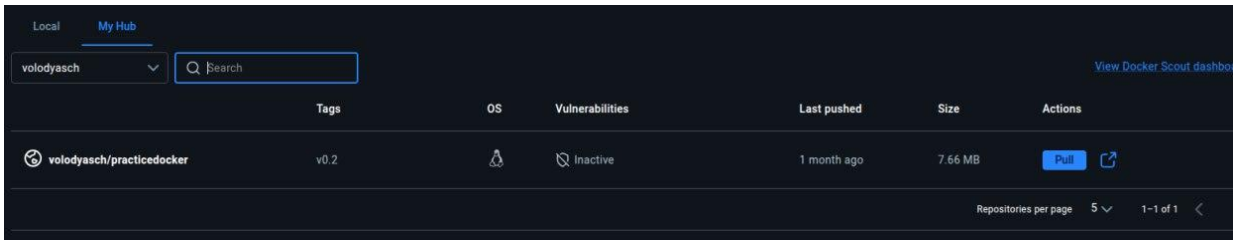
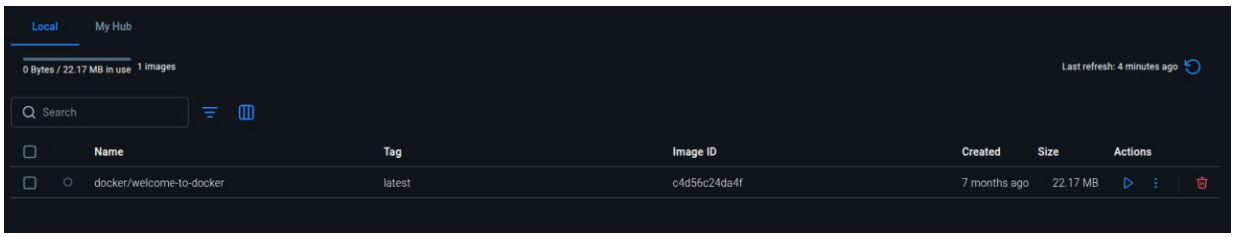

```
admin@Dermen:~/Загрузки$ sudo apt-get update
sudo apt install ./docker-desktop-amd64.deb
Сущ:1 http://ru.archive.ubuntu.com/ubuntu jammy InRelease
Сущ:2 http://ru.archive.ubuntu.com/ubuntu jammy-updates InRelease
Сущ:3 https://download.docker.com/linux/ubuntu jammy InRelease
Сущ:4 http://ru.archive.ubuntu.com/ubuntu jammy-backports InRelease
Сущ:5 https://dl.google.com/linux/chrome/deb stable InRelease
Сущ:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Чтение списков пакетов… Готово
N: Пропускается получение настроенного файла «stable/binary-i386/Packages», так как репозиторий «http
om/linux/ubuntu jammy InRelease» не поддерживает архитектуру «i386»
Чтение списков пакетов… Готово
Построение дерева зависимостей… Готово
Чтение информации о состоянии… Готово
Заметьте, вместо «./docker-desktop-amd64.deb» выбирается «docker-desktop»
Следующие пакеты устанавливались автоматически и больше не требуются:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1 nvidia-firmware-535-535.154.05 nvidia-firmware-535-535.274.02
Для их удаления используйте «sudo apt autoremove».
Будут установлены следующие дополнительные пакеты:
```



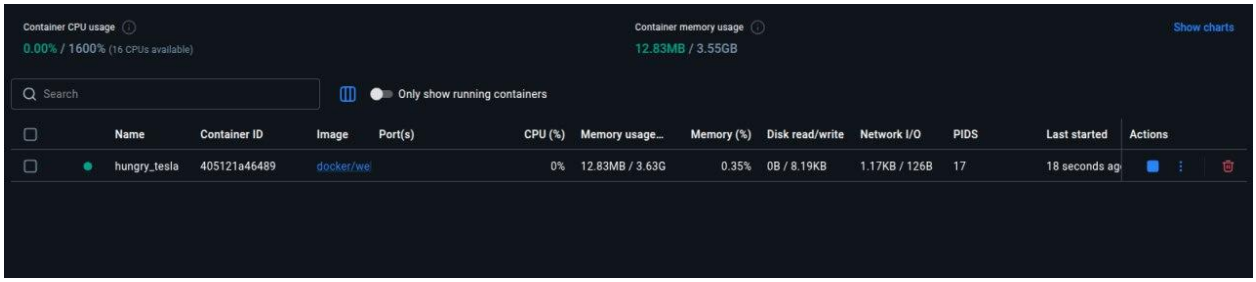

```
1981  /opt/docker-desktop/bin/docker-desktop --version
1982  ls /usr/share/applications | grep docker
1983  nano ~/.local/share/applications/docker-desktop.desktop
1984  chmod +x ~/.local/share/applications/docker-desktop.desktop
1985  update-desktop-database ~/.local/share/applications
```

2.

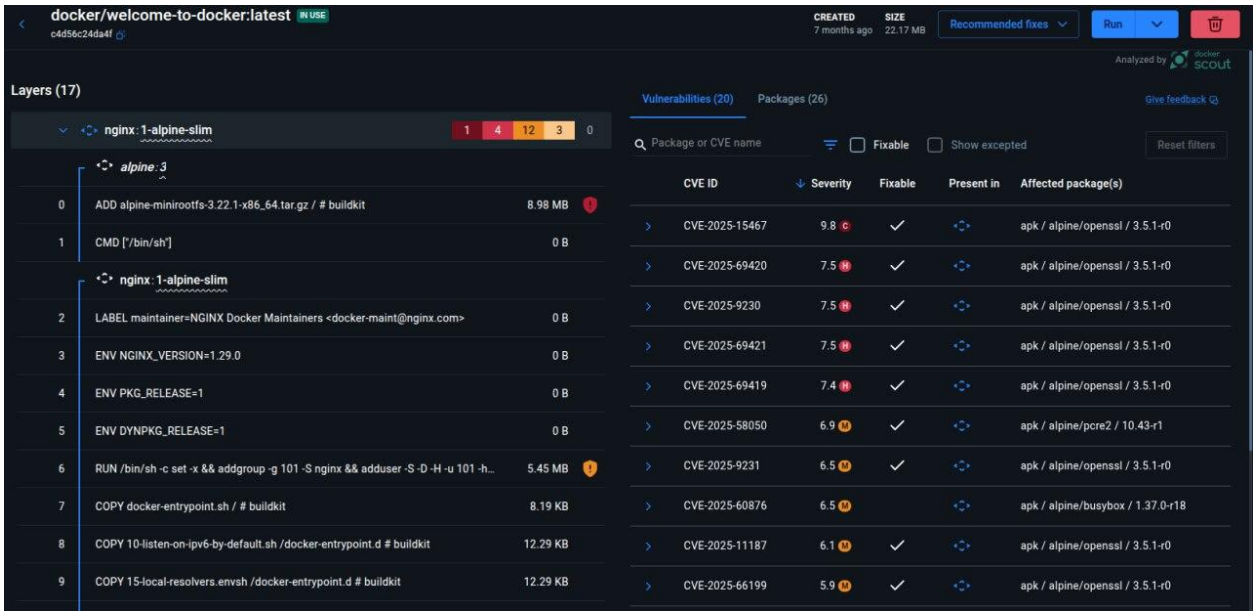



Local
My Hub
0 Bytes / 22.17 MB in use
1 images
Last refresh: 4 minutes ago
Search
Name
Tag
Image ID
Created
Size
Actions
docker/welcome-to-docker
latest
c4d56c24da4f
7 months ago
22.17 MB

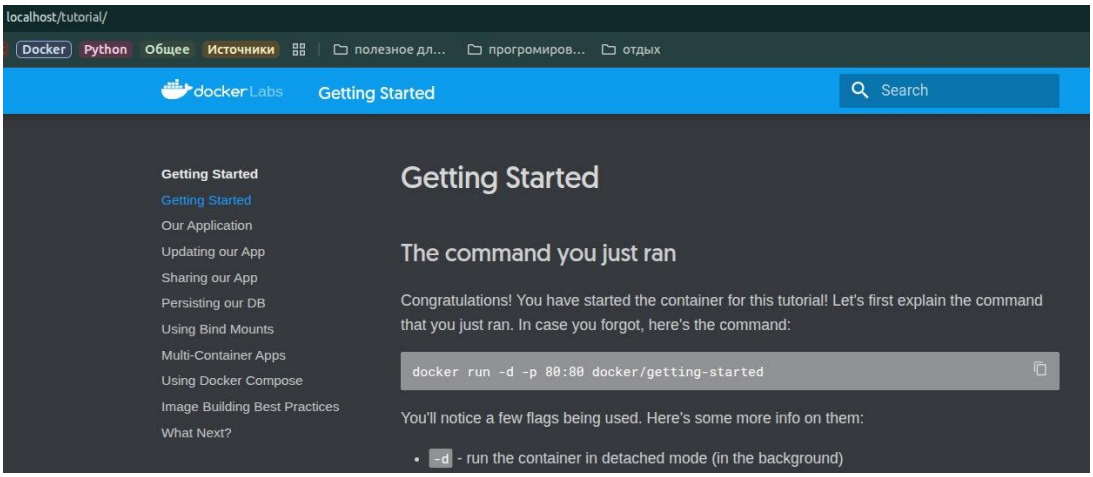
hungry_tesla
405121a46489
docker/welcome-to-docker:latest
STATUS
Running (0 seconds ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/26 22:59:26 [notice] 1#1: using the "epoll" event method
2026/02/26 22:59:26 [notice] 1#1: nginx/1.29.0


Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
12.83MB / 3.55GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Memory usage...
Memory (%)
Disk read/write
Network I/O
PIDS
Last started
Actions
hungry_tesla
405121a46489
docker/wel
0%
12.83MB / 3.63G
0.35%
0B / 8.19KB
1.17KB / 126B
17
18 seconds ag

hungry_tesla
405121a46489
docker/welcome-to-docker:latest
STATUS
Running (47 seconds ago)
Raw JSON
Logs
Inspect
Bind mounts
Exec
Files
Stats
Platform
Cmd
State
Image
PortBindings
Runtime
Mounts
Volumes
Env
Labels
Networks
{
"Id": "405121a464893bf1d8cf3d25f0782d10ccaedb1f7e46d0f4b147a6edf4b231da",
"Created": "2026-02-26T22:59:25.872480897Z",
"Path": "/docker-entrypoint.sh",
"Args": [
"nginx",
"-g",
"daemon off;"
],


docker/welcome-to-docker:latest
IN USE
c4d56c24da4f
CREATED
7 months ago
SIZE
22.17 MB
Recommended fixes
Run
Analyzed by docker scout
Layers (17)
nginx:1-alpine-slim
1 4 12 3 0
alpine:3
0 ADD alpine-minirootfs-3.22.1-x86_64.tar.gz / # buildkit 8.98 MB
1 CMD ['/bin/sh']
nginx:1-alpine-slim
2 LABEL maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>
3 ENV NGINX_VERSION=1.29.0
4 ENV PKG_RELEASE=1
5 ENV DYNPKG_RELEASE=1
6 RUN /bin/sh -c set -x && addgroup -g 101 -S nginx && adduser -S -D -H -u 101 -h... 5.45 MB
7 COPY docker-entrypoint.sh / # buildkit 8.19 KB
8 COPY 10-listen-on-ipv6-by-default.sh /docker-entrypoint.d # buildkit 12.29 KB
9 COPY 15-local-resolvers.envsh /docker-entrypoint.d # buildkit 12.29 KB
Vulnerabilities (20)
Packages (26)
Give feedback
Package or CVE name
Fixable
Show excepted
Reset filters
CVE ID
Severity
Fixable
Present in
Affected package(s)
CVE-2025-15467 9.8 C apk / alpine/openssl / 3.5.1-r0
CVE-2025-69420 7.5 H apk / alpine/openssl / 3.5.1-r0
CVE-2025-9230 7.5 H apk / alpine/openssl / 3.5.1-r0
CVE-2025-69421 7.5 H apk / alpine/openssl / 3.5.1-r0
CVE-2025-69419 7.4 H apk / alpine/openssl / 3.5.1-r0
CVE-2025-58050 6.9 M apk / alpine/pcre2 / 10.43-r1
CVE-2025-9231 6.5 M apk / alpine/openssl / 3.5.1-r0
CVE-2025-60876 6.5 M apk / alpine/busybox / 1.37.0-r18
CVE-2025-11187 6.1 M apk / alpine/openssl / 3.5.1-r0
CVE-2025-66199 5.9 M apk / alpine/openssl / 3.5.1-r0

```
admin@Dermen:~/Docker/Docker_Tutor$ docker pull docker/getting-started
Using default tag: latest
latest: Pulling from docker/getting-started
b6334b6ace34: Pull complete
33e0cbbb4673: Pull complete
1e35f6679fab: Pull complete
4f7e34c2de10: Pull complete
cb9626c74200: Pull complete
f1d1c9928c82: Pull complete
c158987b0551: Pull complete
ee68d3549ec8: Pull complete
9b6f639ec6ea: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
docker.io/docker/getting-started:latest
```




```
admin@Dermen:~/Docker/Docker_Tutor$ cd app
admin@Dermen:~/Docker/Docker_Tutor/app$ ls
Dockerfile  package.json  spec  src  yarn.lock
admin@Dermen:~/Docker/Docker_Tutor/app$ vi Dockerfile
admin@Dermen:~/Docker/Docker_Tutor/app$ cat Dockerfile
FROM node:18-alpine
WORKDIR /app
COPY ..
RUN yarn install --production
CMD ["node", "src/index.js"]
admin@Dermen:~/Docker/Docker_Tutor/app$ docker build -t getting-started
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
admin@Dermen:~/Docker/Docker_Tutor/app$ docker build -t getting-started .
[+] Building 0.1s (1/1) FINISHED                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                      0.0s
 => => transferring dockerfile: 137B                                      0.0s
```

3.

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2$ git clone https://github.com/smartiqaorg/geometric_lib
Клонирование в «geometric_lib»...
remote: Enumerating objects: 35, done.
remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)
Получение объектов: 100% (35/35), 4.69 КиБ | 4.69 МиБ/с, готово.
Определение изменений: 100% (5/5), готово.
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2$ tree
.
├── command
└── geometric_lib
    ├── circle.py
    ├── docs
    │   └── README.md
    └── square.py
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ echo "" > config.json
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ ls
circle.py  config.json  docs  square.py
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ vi config.json
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ > Dockerfile
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ vi Dockerfile
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ docker build -t geometric-app .
[+] Building 3.9s (9/9) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 258B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                 0.2s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.12-slim@sha256:f3fa41d74a768c2fce8016b98c191ae8c1bacd8f1152870a3f9f87d350920b7c  0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:f3fa41d74a768c2fce8016b98c191ae8c1bacd8f1152870a3f9f87d350920b7c  0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 402B                                                                                   0.0s
 => CACHED [2/4] WORKDIR /app                                                                                       0.0s
 => CACHED [3/4] RUN apt-get update && apt-get install -y git && rm -rf /var/lib/apt/lists/*                        0.0s
```

Dockerfile

```
FROM python:3.12-slim

WORKDIR /app

COPY circle.py square.py main.py .

CMD ["python", "main.py"]
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$ docker run --rm geometry-app
Circle (r=5)
Area: 78.53981633974483
Perimeter: 31.41592653589793

Square (a=6)
Area: 36
Perimeter: 24
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/geometric_lib$
```

4-6.

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2$ git clone https://github.com/bradtraversy/proshop_mern.git
Клонирование в «proshop_mern»...
remote: Enumerating objects: 901, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 901 (delta 2), reused 1 (delta 1), pack-reused 897 (from 2)
Получение объектов: 100% (901/901), 1.20 МиБ | 1.60 МиБ/с, готово.
Определение изменений: 100% (607/607), готово.
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2$ ls
command  geometric_lib  proshop_mern
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$  ls
backend  frontend  package.json  package-lock.json  Procfile  README.md  uploads
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ cd backend
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/backend$ > DOckerfile
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/backend$ vi Dockerfile
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/backend$ cd ..
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ cd frontend
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/frontend$ > Dockerfile
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/frontend$ vi Dockerfile
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/frontend$ cd ..
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ >docker-compose.yml
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ vi docker-compose.yml
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker compose -f docker-compose.dev.yml up --build
WARN[0000] /home/karina/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern/docker-compose.dev.yml: the attribute `version`
is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 8/10
 ⠿ Image mongo:4.2 [⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿] 162.9MB / 164.3MB Pulling                                          16.5s
[+] Building 50.9s (20/20) FINISHED
 => [internal] load local bake definitions                                                                          0.0s
 => => reading from stdin 1.10kB                                                                                    0.0s
 => [frontend internal] load build definition from Dockerfile                                                       0.0s
 => => transferring dockerfile: 183B                                                                                0.0s
 => [server internal] load build definition from Dockerfile                                                         0.0s
 => => transferring dockerfile: 202B                                                                                0.0s
```

```yaml
services:
  mongo:
    image: mongo:4.2
    container_name: proshop_mongo
    restart: always
    ports:
      - "27018:27017"
    volumes:
      - mongo_data:/data/db

  server:
    build:
      context: .
      dockerfile: backend/Dockerfile
    container_name: proshop_server
    restart: always
    environment:
      - NODE_ENV=development
      - PORT=5000
      - MONGO_URI=mongodb://mongo:27017/proshop
      - JWT_SECRET=abc123
    ports:
      - "5000:5000"
    volumes:
      - ./:/usr/src/app
      - /usr/src/app/node_modules
    depends_on:
      - mongo

  frontend:
    build:
      context: .
      dockerfile: frontend/Dockerfile
    container_name: proshop_frontend
    environment:
      - CHOKIDAR_USEPOLLING=true
    ports:
      - "3000:3000"
    volumes:
      - ./frontend:/usr/src/app
      - /usr/src/app/node_modules
    stdin_open: true
    tty: true
    depends_on:
      - server

volumes:
  mongo_data:
```

```
FROM node:18-alpine

WORKDIR /usr/src/app

COPY package*.json ./
RUN npm install

COPY . .

EXPOSE 5000

CMD npm run server
```

```
FROM node:18-alpine
WORKDIR /usr/src/app

ENV NODE_OPTIONS=--openssl-legacy-provider
ENV PATH /usr/src/app/node_modules/.bin:$PATH

COPY frontend/package*.json ./
RUN npm ci
COPY frontend/ ./

EXPOSE 3000
CMD ["npm", "start"]
```

```
docker tag proshop_mern-frontend:latest volodyasch/proshop-frontend:latest
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker images
                                                                                    i Info →  U  In Use
IMAGE                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/getting-started:latest           d79336f4812b       73.9MB         21.4MB    U
geometric-app:latest                    ab97806eb2c1        177MB         43.2MB    U
mongo:4.2                               699d652ed674        568MB          166MB    U
myusername/proshop_frontend:latest      680585ff0f94        847MB          165MB    U
myusername/proshop_server:latest        fef7405ad6b6        300MB           60MB    U
prizz/deep-dive:1.0.95                  7b310e76b649       49.5MB         14.4MB    U
proshop_mern-frontend:latest            593063b727fe        847MB          165MB    U
proshop_mern-server:latest              7438e983fc63        300MB           60MB    U
volodyasch/proshop-frontend:latest      593063b727fe        847MB          165MB    U
volodyasch/proshop-server:latest        7438e983fc63        300MB           60MB    U
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker push myusername/proshop-server:latest
docker push myusername/proshop-frontend:latest
The push refers to repository [docker.io/myusername/proshop-server]
tag does not exist: myusername/proshop-server:latest
The push refers to repository [docker.io/myusername/proshop-frontend]
tag does not exist: myusername/proshop-frontend:latest
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$
```

```
admin@Dermen:~$ docker push volodyasch/proshop-server:latest
The push refers to repository [docker.io/volodyasch/proshop-server]
1e5a4c89cee5: Layer already exists
368d34f0ea25: Layer already exists
25ff2da83641: Layer already exists
6041081bb73d: Pushed
dd71dde834b5: Pushed
758cbb9be1a2: Layer already exists
0880b066cbcb: Layer already exists
f18232174bc9: Layer already exists
957bcb29f665: Already exists
latest: digest: sha256:7438e983fc63d697fff77180ff3ea6458ba6f7fa1cd3c723dfeb12bba
caff9ab size: 856
```

7.

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
d3caac714a7f   host                                      host      local
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "2ffc5be88a094fafbc6089aa5132a1876c4fb9cf24b1e28cb6465b3e4788a81e",
        "Created": "2026-03-01T23:00:48.964047626Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network create mu_first_bridg
4d34d74fbb99ac3293cdbc00652df157984fe41ab5c889a8ca5672723396f406
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
d3caac714a7f   host                                      host      local
4d34d74fbb99   mu_first_bridg                            bridge    local
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect mu_first_bridg
[
    {
        "Name": "mu_first_bridg",
        "Id": "4d34d74fbb99ac3293cdbc00652df157984fe41ab5c889a8ca5672723396f406",
        "Created": "2026-03-02T00:04:17.27177696Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker run -dit --name test_container --network mu_first_bridg alpine sh
d865a1365d108dccf56f4afb54f166524bc1d080a13ad08444fcdacbd34f39a9
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED             STATUS              PORTS
d865a1365d10   alpine                 "sh"                     10 seconds ago      Up 9 seconds
45cd3f2c218b   proshop_mern-frontend  "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:3000->3000/tcp, [::]:3000-
4e6dee4f77ee   proshop_mern-server    "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:5000->5000/tcp, [::]:5000-
586a0a1906e6   mongo:4.2              "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:27018->27017/tcp, [::]:270
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect mu_first_bridg
[
    {
        "Name": "mu_first_bridg",
        "Id": "4d34d74fbb99ac3293cdbc00652df157984fe41ab5c889a8ca5672723396f406",
        "Created": "2026-03-02T00:04:17.27177696Z",
        "Scope": "local",
        "Driver": "bridge",
```

```
    },
    "Labels": {},
    "Containers": {
        "d865a1365d108dccf56f4afb54f166524bc1d080a13ad08444fcdacbd34f39a9": {
            "Name": "test_container",
            "EndpointID": "ec86f78939972d1dc66afde7bf181b44e48c73fe17e815ad006f39301b89f238",
            "MacAddress": "a6:48:45:eb:4e:a8",
            "IPv4Address": "172.19.0.2/16",
            "IPv6Address": ""
        }
    },
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network disconnect mu_first_bridg test_container
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker ps
CONTAINER ID   IMAGE                   COMMAND                  CREATED             STATUS              PORTS
d865a1365d10   alpine                  "sh"                     4 minutes ago       Up 4 minutes
45cd3f2c218b   proshop_mern-frontend   "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:3000->306
4e6dee4f77ee   proshop_mern-server     "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:5000->506
586a0a1906e6   mongo:4.2               "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:27018->27
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect mu_first_bridg
[
```

```
        },
        "Labels": {},
        "Containers": {},
        "Status": {
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network create --driver bridge my_bridge
docker network ls
docker network inspect my_bridge2
aeda190e54e61383eb742101fd338a508ed0ff0c5bf3df1f4fb845c9501072fa
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
d3caac714a7f   host                                      host      local
4d34d74fbb99   mu_first_bridg                            bridge    local
aeda190e54e6   my_bridge2                                bridge    local
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
[
    {
        "Name": "my_bridge2",
        "Id": "aeda190e54e61383eb742101fd338a508ed0ff0c5bf3df1f4fb845c9501072fa",
        "Created": "2026-03-02T00:14:48.612161992Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.21.0.0/16",
                    "Gateway": "172.21.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
```

```
Run 'docker run --help' for more information
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker run -dit --name container1 --network my_bridge2 alpine sh
acd57fb2ec32a8799fd7e1b62b567c1540fe45eb122ac1e9d7c8ed149fb0939c
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker run -dit --name container2 --network my_bridge2 alpine sh
0faecc5d0d78f6743f374c345a83ae3e0ae1a16bf97f03314a0dcc811f814818
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker run -dit --name container3 --network my_bridge2 alpine sh
990daf4858bf1ebfbaf701ed3f14a49e4f998f2d72c0488531268511cc629e70
```

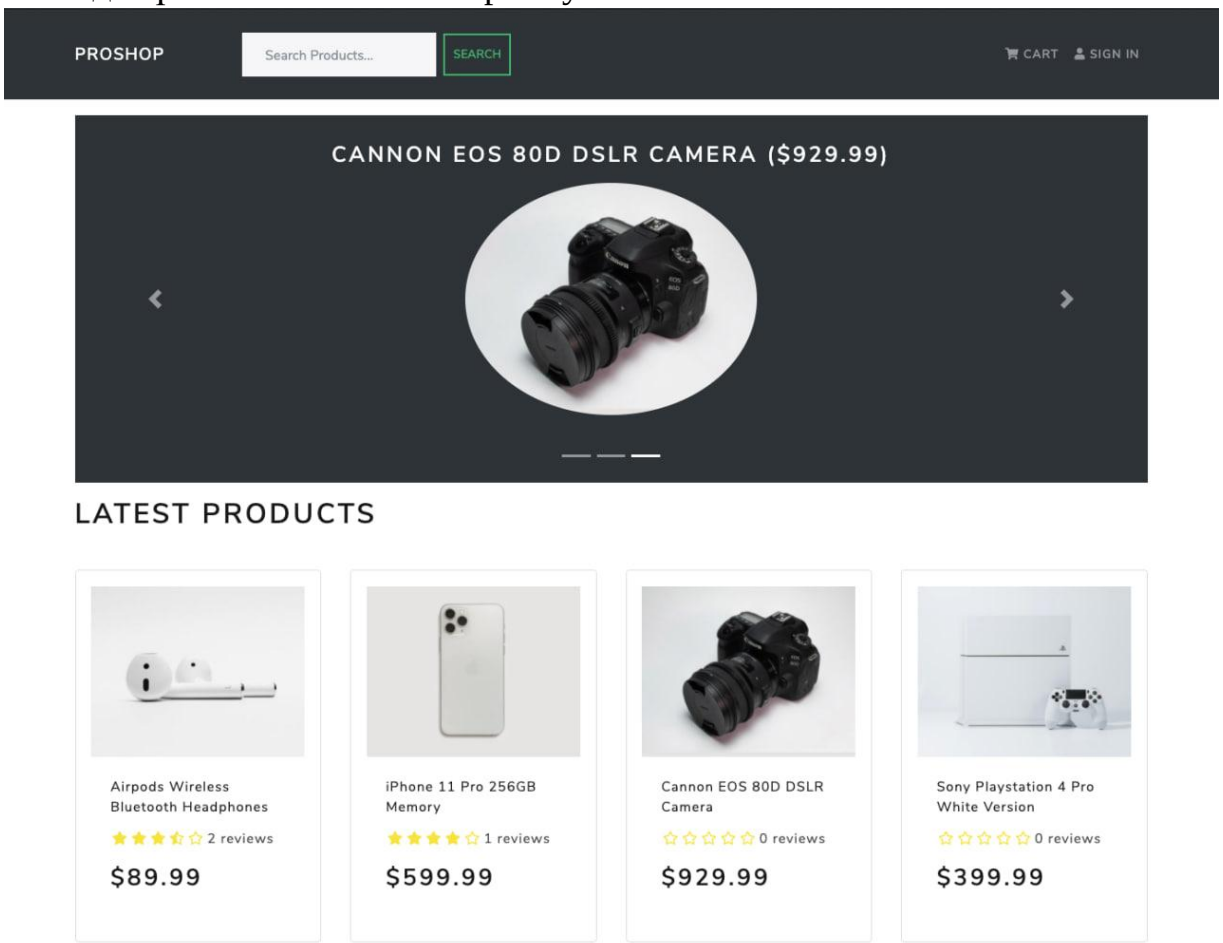
```
/ # ping c1
ping: bad address 'c1'
/ # ping container2
PING container2 (172.21.0.3): 56 data bytes
64 bytes from 172.21.0.3: seq=0 ttl=64 time=0.144 ms
64 bytes from 172.21.0.3: seq=1 ttl=64 time=0.241 ms
64 bytes from 172.21.0.3: seq=2 ttl=64 time=0.255 ms
64 bytes from 172.21.0.3: seq=3 ttl=64 time=0.212 ms
64 bytes from 172.21.0.3: seq=4 ttl=64 time=0.207 ms
64 bytes from 172.21.0.3: seq=5 ttl=64 time=0.212 ms
64 bytes from 172.21.0.3: seq=6 ttl=64 time=0.213 ms
64 bytes from 172.21.0.3: seq=7 ttl=64 time=0.168 ms
64 bytes from 172.21.0.3: seq=8 ttl=64 time=0.215 ms
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker exec -it container1 sh
/ # ping -c 3 container2
PING container2 (172.21.0.3): 56 data bytes
64 bytes from 172.21.0.3: seq=0 ttl=64 time=0.146 ms
64 bytes from 172.21.0.3: seq=1 ttl=64 time=0.238 ms
64 bytes from 172.21.0.3: seq=2 ttl=64 time=0.241 ms

--- container2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.146/0.208/0.241 ms
/ #
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker swarm init
Swarm initialized: current node (mkf5a3dofbl9s506met652932) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-3eyci7k5biwivxahman59i2q3g6kntmul5p0fyyzir11d3c7c0-8lhl5gsm2d1gmf

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network create -d overlay my_overl
l8h2lfolo8pnwse8dthujohaf
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
ae0943db0b71   docker_gwbridge                           bridge    local
d3caac714a7f   host                                      host      local
kgmpdkbn87ad   ingress                                   overlay   swarm
4d34d74fbb99   mu_first_bridg                            bridge    local
aeda190e54e6   my_bridge2                                bridge    local
l8h2lfolo8pn   my_overlay                                overlay   swarm
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect my_overlay
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network create -d overlay my_overlay2
6w90efmtinyp0osvzfz3978a3
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network inspect my_overlay2
[
    {
        "Name": "my_overlay2",
        "Id": "6w90efmtinyp0osvzfz3978a3",
        "Created": "2026-03-02T00:32:28.85581022Z",
        "Scope": "swarm",
        "Driver": "overlay",
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
ae0943db0b71   docker_gwbridge                           bridge    local
d3caac714a7f   host                                      host      local
kgmpdkbn87ad   ingress                                   overlay   swarm
4d34d74fbb99   mu_first_bridg                            bridge    local
aeda190e54e6   my_bridge2                                bridge    local
l8h2lfolo8pn   my_overlay                                overlay   swarm
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$
```

```
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network create -d host my_host
Error response from daemon: only one instance of "host" network is allowed
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$ docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
2ffc5be88a09   bridge                                    bridge    local
ae0943db0b71   docker_gwbridge                           bridge    local
d3caac714a7f   host                                      host      local
kgmpdkbn87ad   ingress                                   overlay   swarm
4d34d74fbb99   mu_first_bridg                            bridge    local
aeda190e54e6   my_bridge2                                bridge    local
l8h2lfolo8pn   my_overlay                                overlay   swarm
b6cd84b5a135   none                                      null      local
4313432dbf24   prizz_deep-dive-desktop-extension_default bridge    local
f20375e5e184   proshop_mern_default                      bridge    local
admin@Dermen:~/453505/453505_VLADIMIR_14/IGI/LR2/proshop_mern$
```

Сайт для работы с docker-compose.yml: